

Markscheme

Mathematics: analysis and approaches

Practice question bank

64. (a)

The sum of the first n terms of an arithmetic sequence is given by $S_n = 4n^2 - n$.

Find the common difference of the sequence.

use $u_n = S_n - S_{n-1}$ to find the general term: $u_n = (4n^2 - n) - (4(n-1)^2 - (n-1))$ (M1)

$$u_n = 8n - 5 \quad \text{A1}$$

since u_n is linear in n with gradient 8, the sequence is arithmetic with (M1)

$$d = 8 \quad \text{A1}$$

[6 marks]

(b)

Hence find the value of u_1 .

$$u_1 = 8(1) - 5 \quad (M1)$$

$$= 3 \quad \text{A1}$$

[2 marks]